

The spinfull ferroaxial model with KM SOC

$$H_{\text{spinfull}} = H_{\text{twobands}} \times s_0 + 2t_\lambda u(k) \times (\sigma_z \times s_z)$$

$$u(k) = \sum_{i=1}^3 \sin(k \cdot \Delta_i) \quad \Delta_i = [\delta_1 - \delta_2, \delta_2 - \delta_3, \delta_3 - \delta_1]$$

$$H_{2\text{bands}} = \begin{pmatrix} -\mu + \Delta + t_2 f_2(k) & t f(k) + t_3 g(k) + t_\perp m(k) \\ h.c. & -\mu - \Delta + t_2 f_2(k) \end{pmatrix}$$

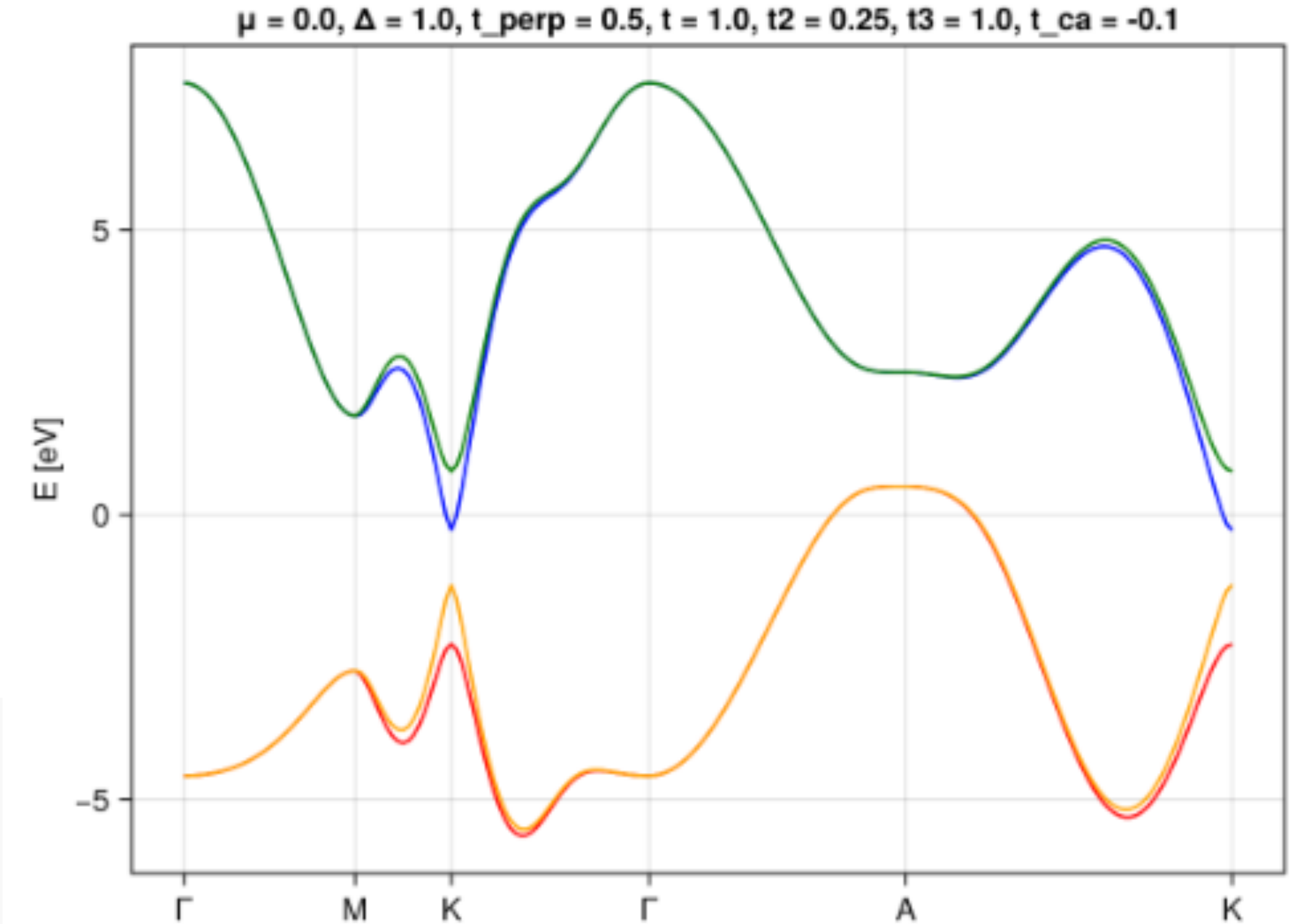
Same as before

1st $f(k) = \sum e^{i\vec{k} \cdot \vec{\delta}_i}$

2nd $f_2(k) = e^{i\vec{k} \cdot (\vec{\delta}_1 - \vec{\delta}_2)} + e^{i\vec{k} \cdot (\vec{\delta}_2 - \vec{\delta}_3)} + e^{i\vec{k} \cdot (\vec{\delta}_3 - \vec{\delta}_1)} + e^{i\vec{k} \cdot (-\vec{\delta}_1 + \vec{\delta}_2)} + e^{i\vec{k} \cdot (-\vec{\delta}_2 + \vec{\delta}_3)} + e^{i\vec{k} \cdot (-\vec{\delta}_3 + \vec{\delta}_1)}$

3rd (opposite signs) $g(\vec{k}) = e^{i\vec{k} \cdot (2\vec{\delta}_1 - \vec{\delta}_2)} + e^{i\vec{k} \cdot (2\vec{\delta}_2 - \vec{\delta}_3)} + e^{i\vec{k} \cdot (2\vec{\delta}_3 - \vec{\delta}_1)} - e^{i\vec{k} \cdot (2\vec{\delta}_1 - \vec{\delta}_3)} - e^{i\vec{k} \cdot (2\vec{\delta}_2 - \vec{\delta}_1)} - e^{i\vec{k} \cdot (2\vec{\delta}_3 - \vec{\delta}_2)}$

Perpendicular $m(k) = \sum e^{i\vec{k} \cdot (\vec{\delta}_i + \vec{a}_3)} + \sum e^{i\vec{k} \cdot (\vec{\delta}_i - \vec{a}_3)}$



Parameters

$$\Delta = 1 \text{ eV}$$

$$t = t_3 = 1 \text{ eV}$$

$$t_\perp = 0.5 \text{ eV}$$

$$t_2 = 0.25 \text{ eV}$$

$$\mu = 0 \text{ eV}$$

$$t_\lambda = -0.1 \text{ eV}$$

$$a_1 = a/2 * [3, \sqrt{3}, 0]$$

$$a_2 = a/2 * [3, -\sqrt{3}, 0]$$

$$a_3 = a * [0, 0, 1]$$

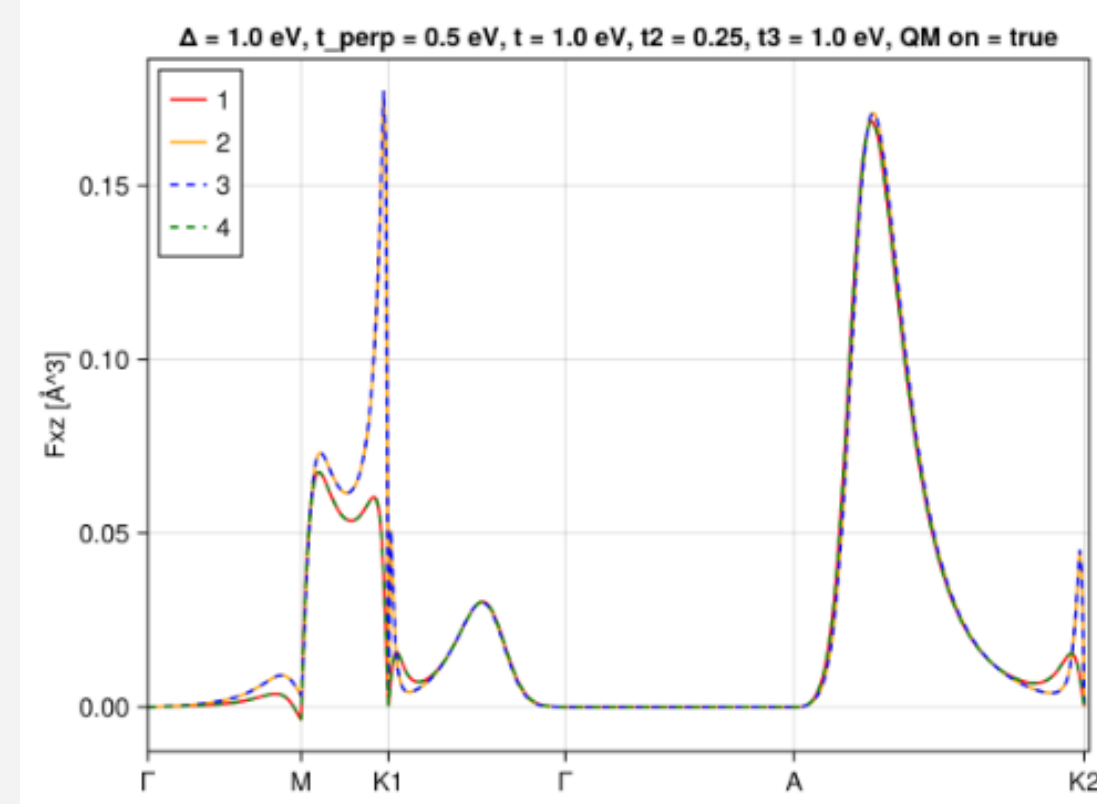
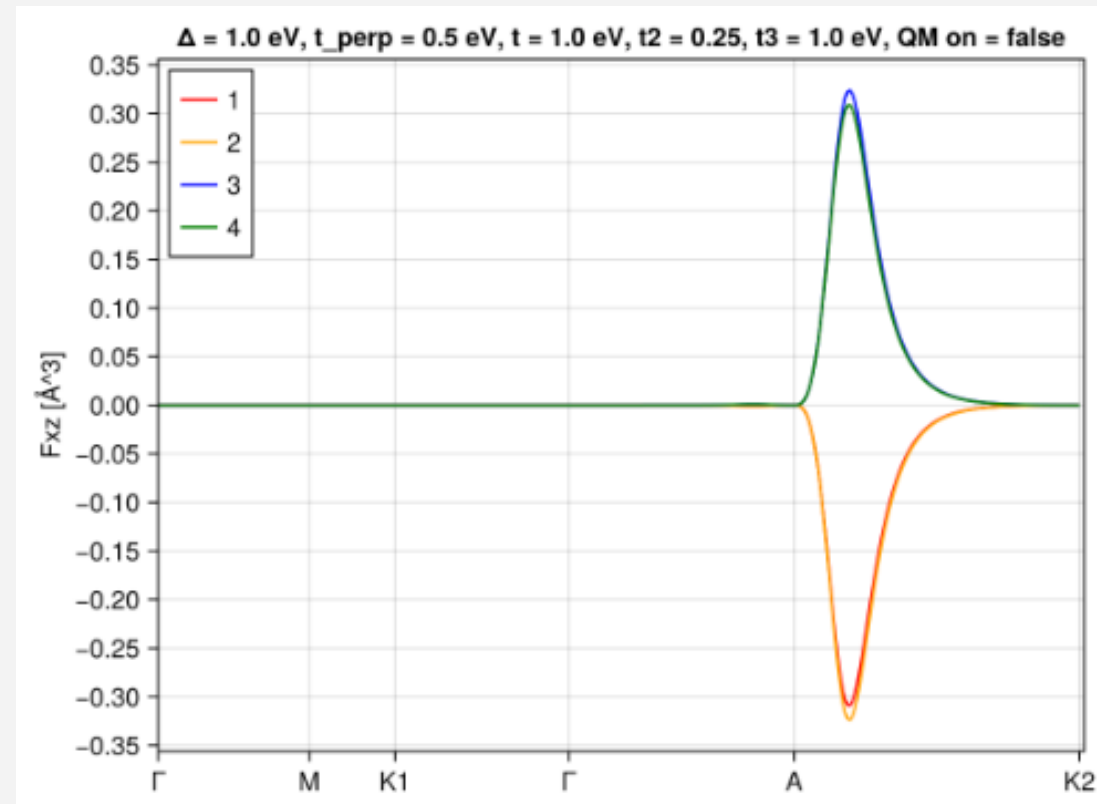
$$a = c = 1 \text{ \AA}$$

K-resolved calculations: F_{ij}

OMM

F_{xz}

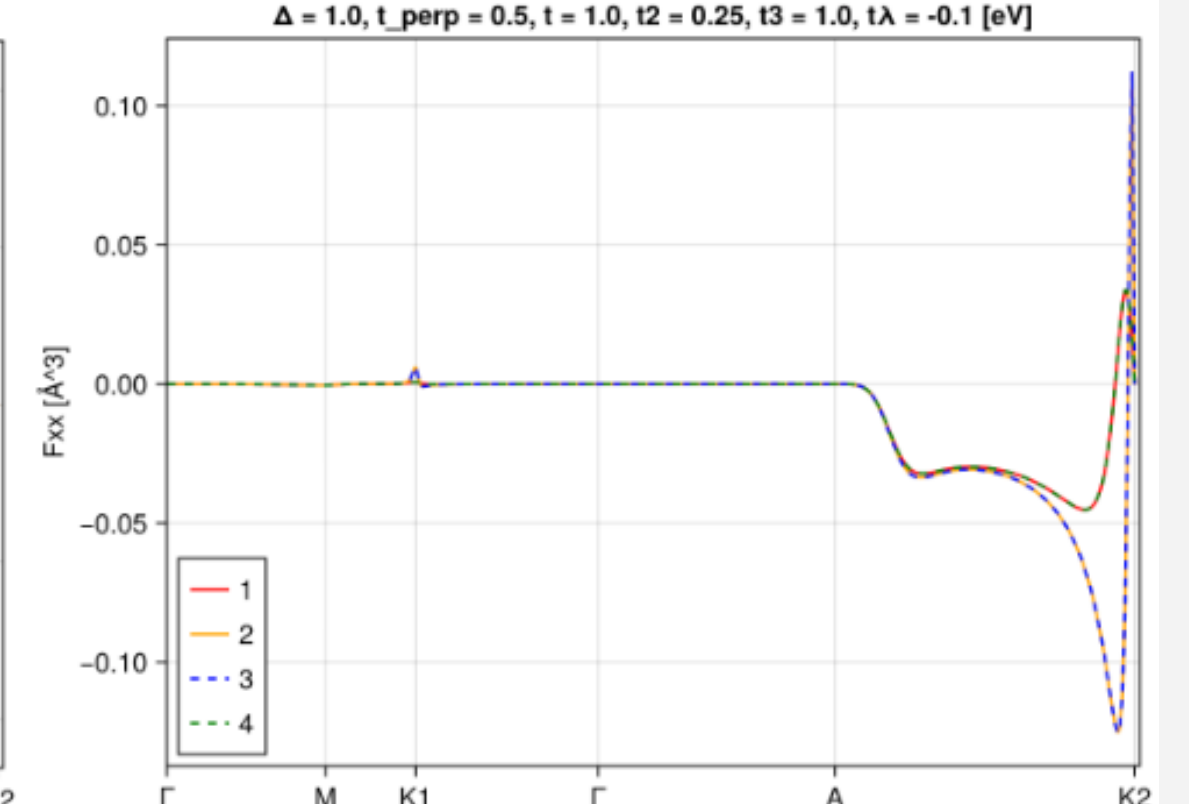
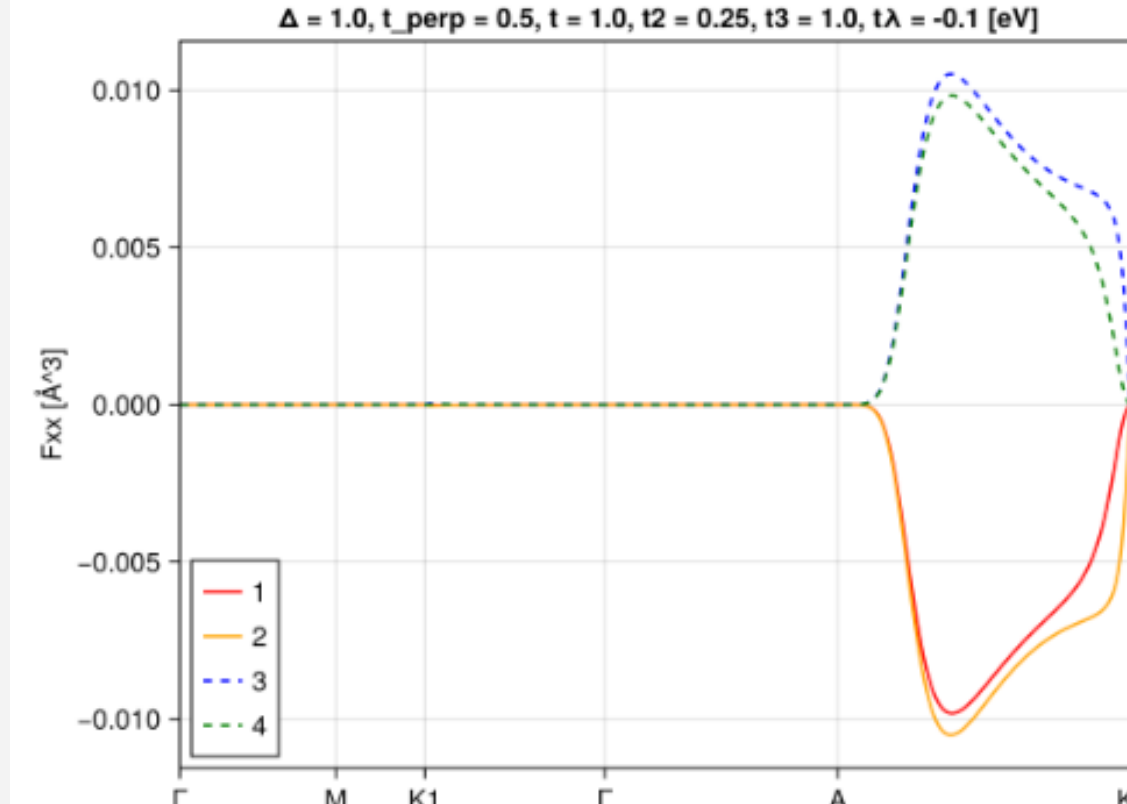
QM



OMM

F_{xx}

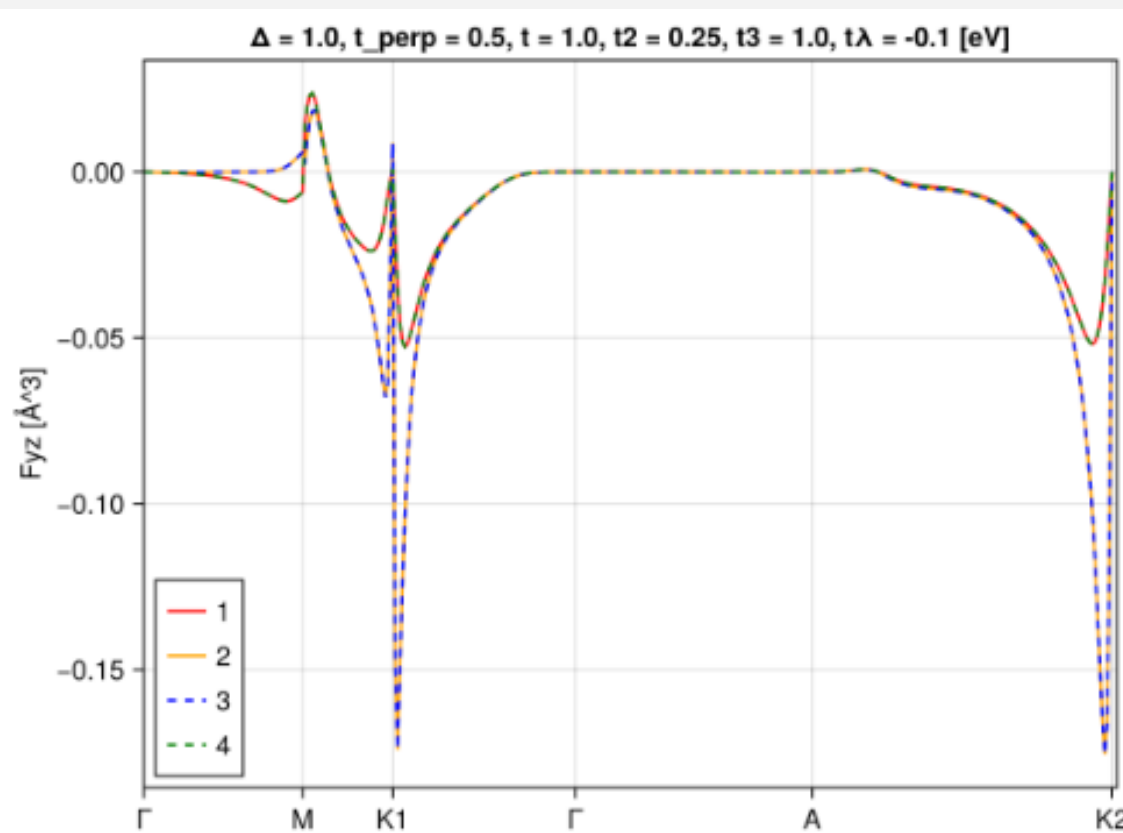
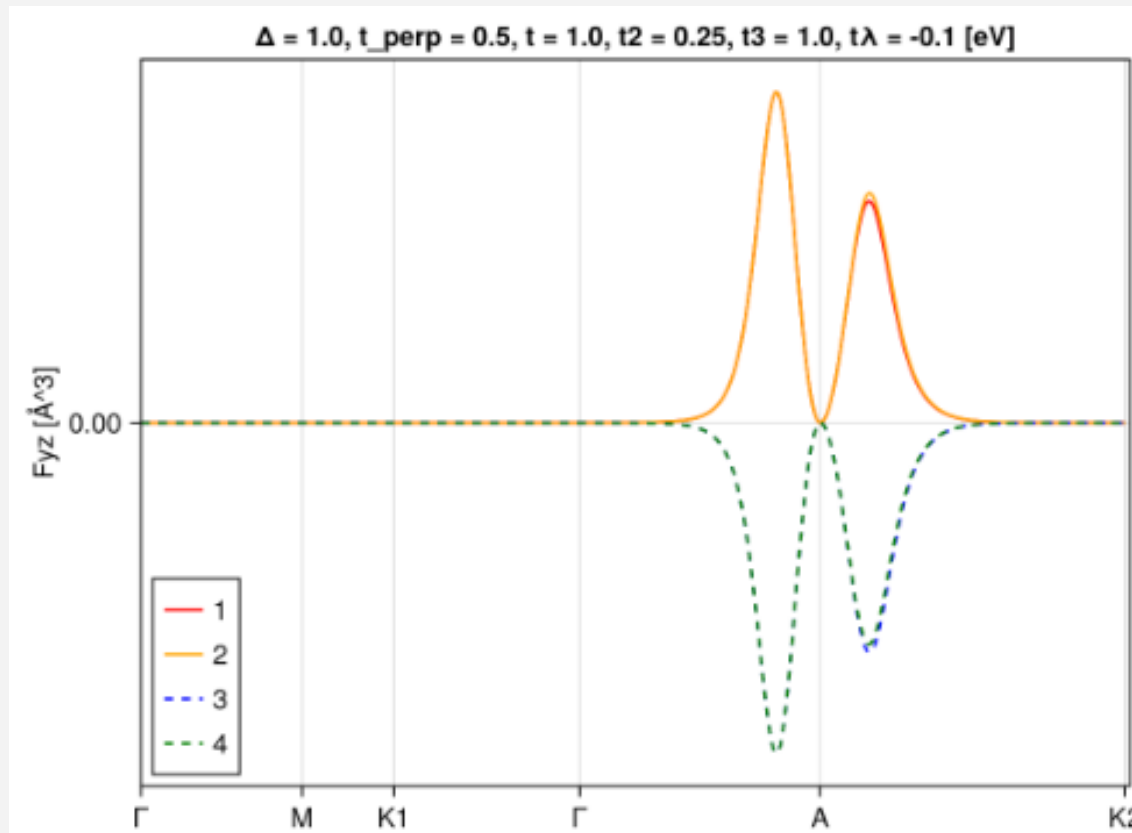
QM



OMM

F_{yz}

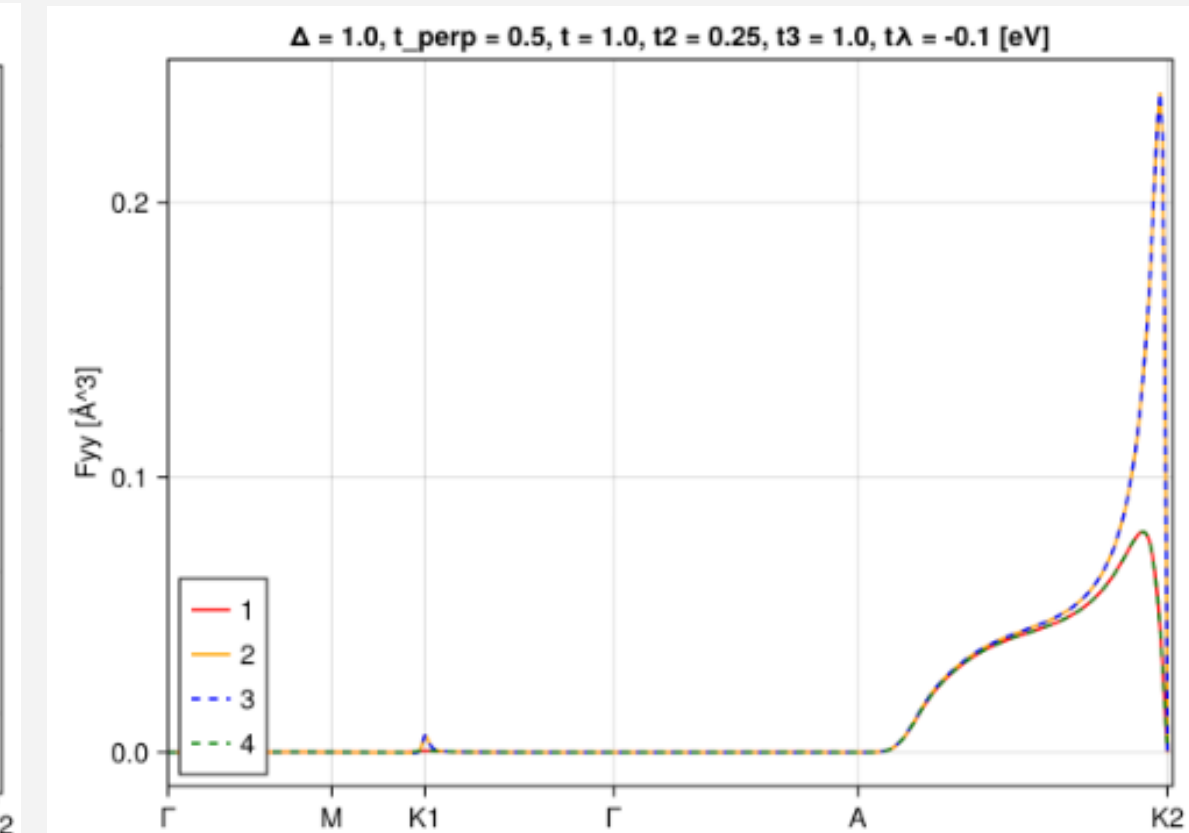
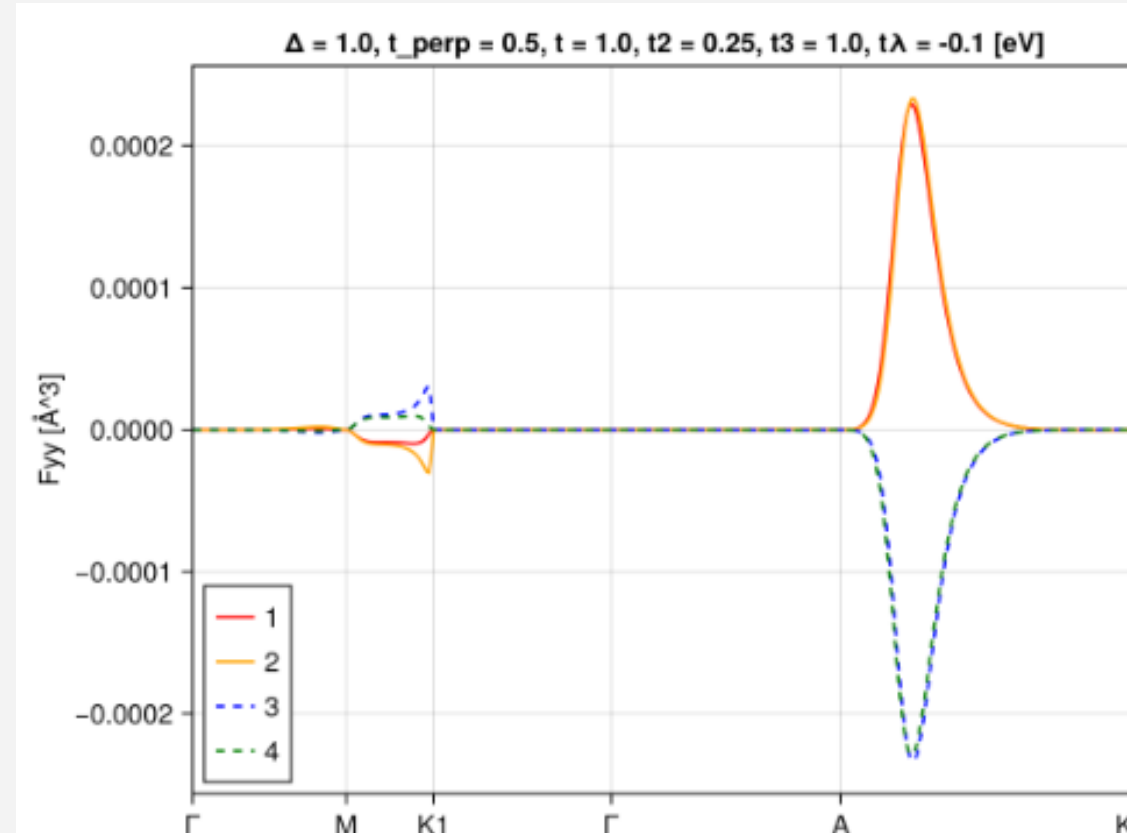
QM



OMM

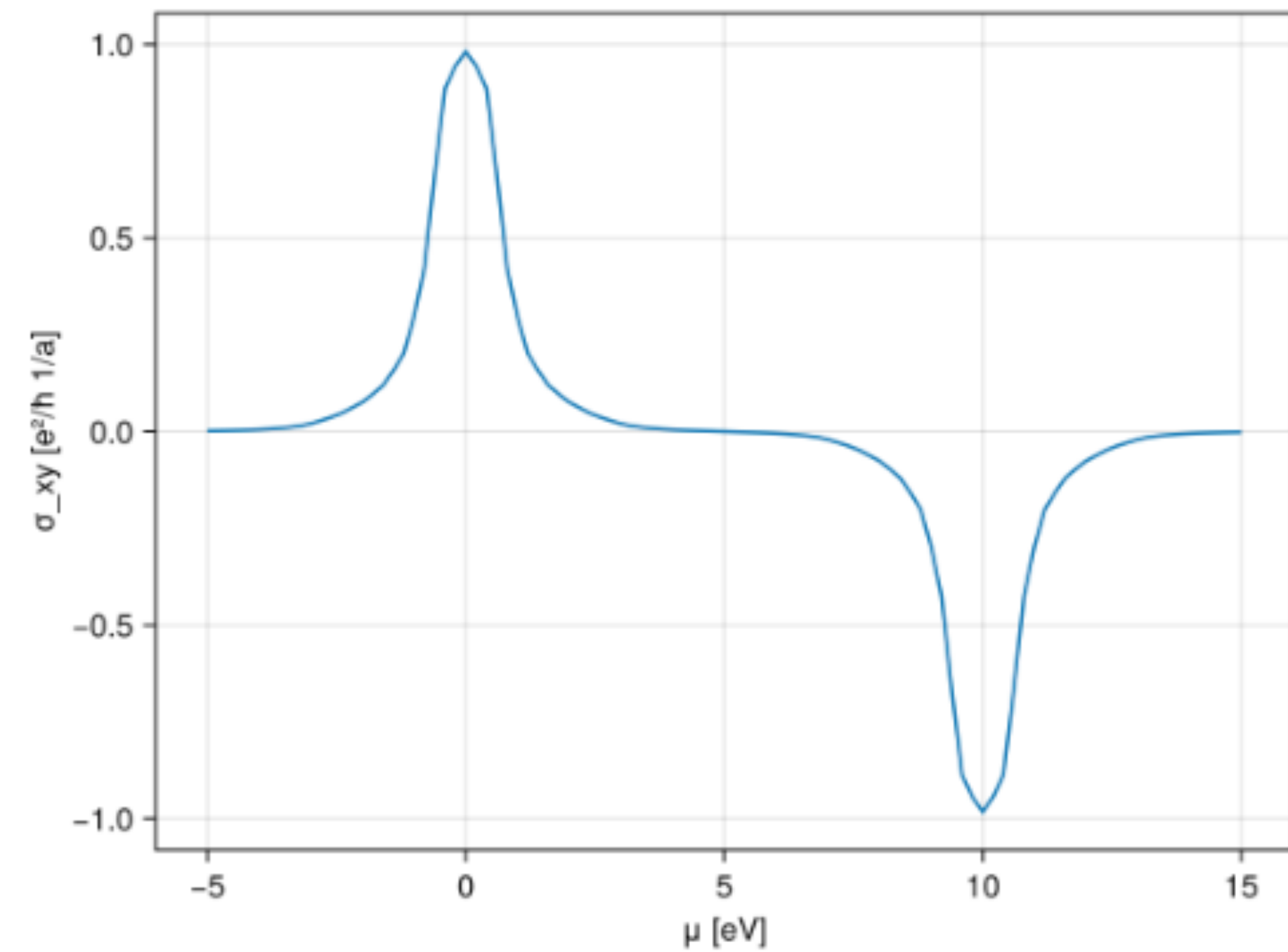
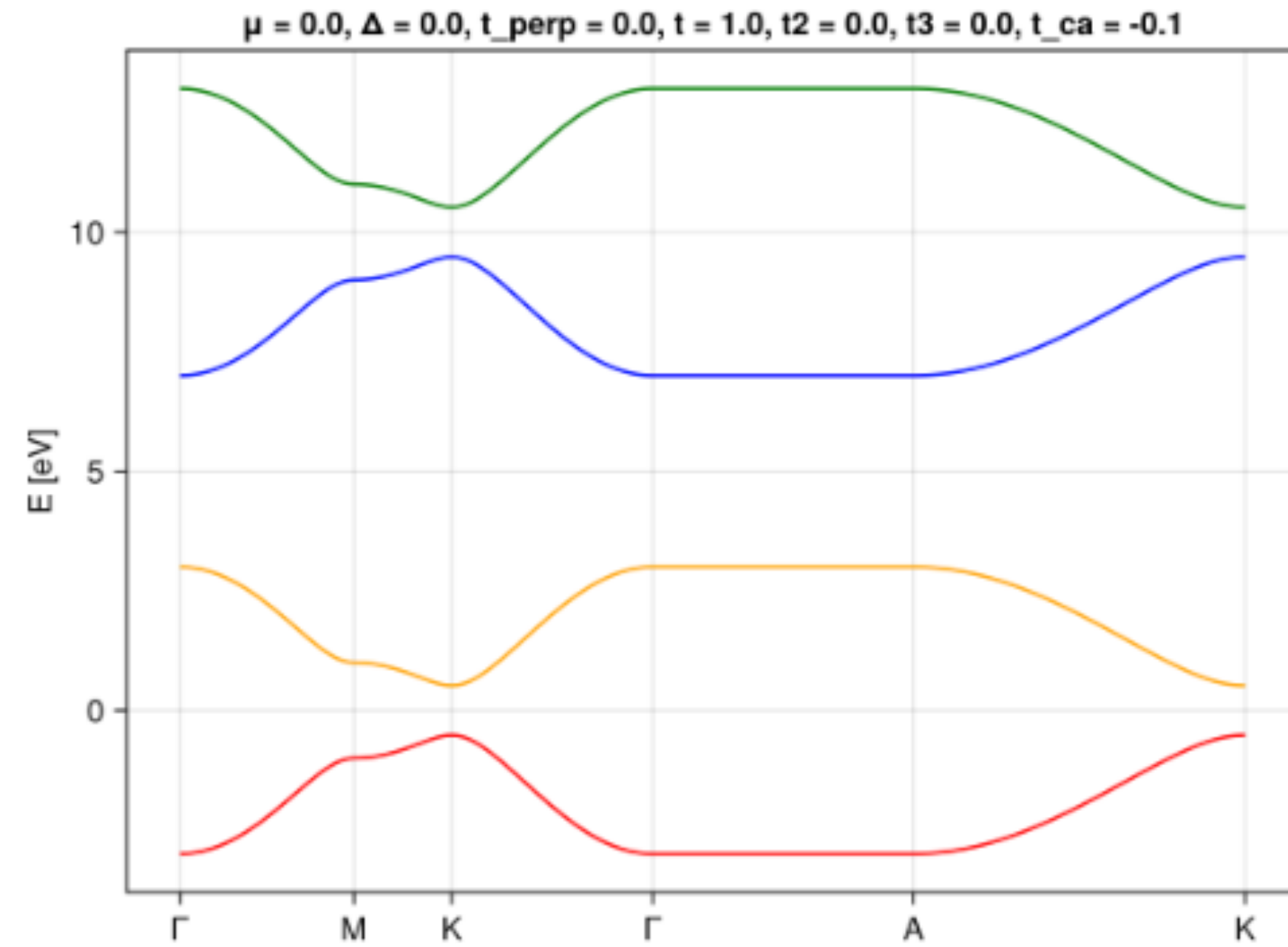
F_{yy}

QM



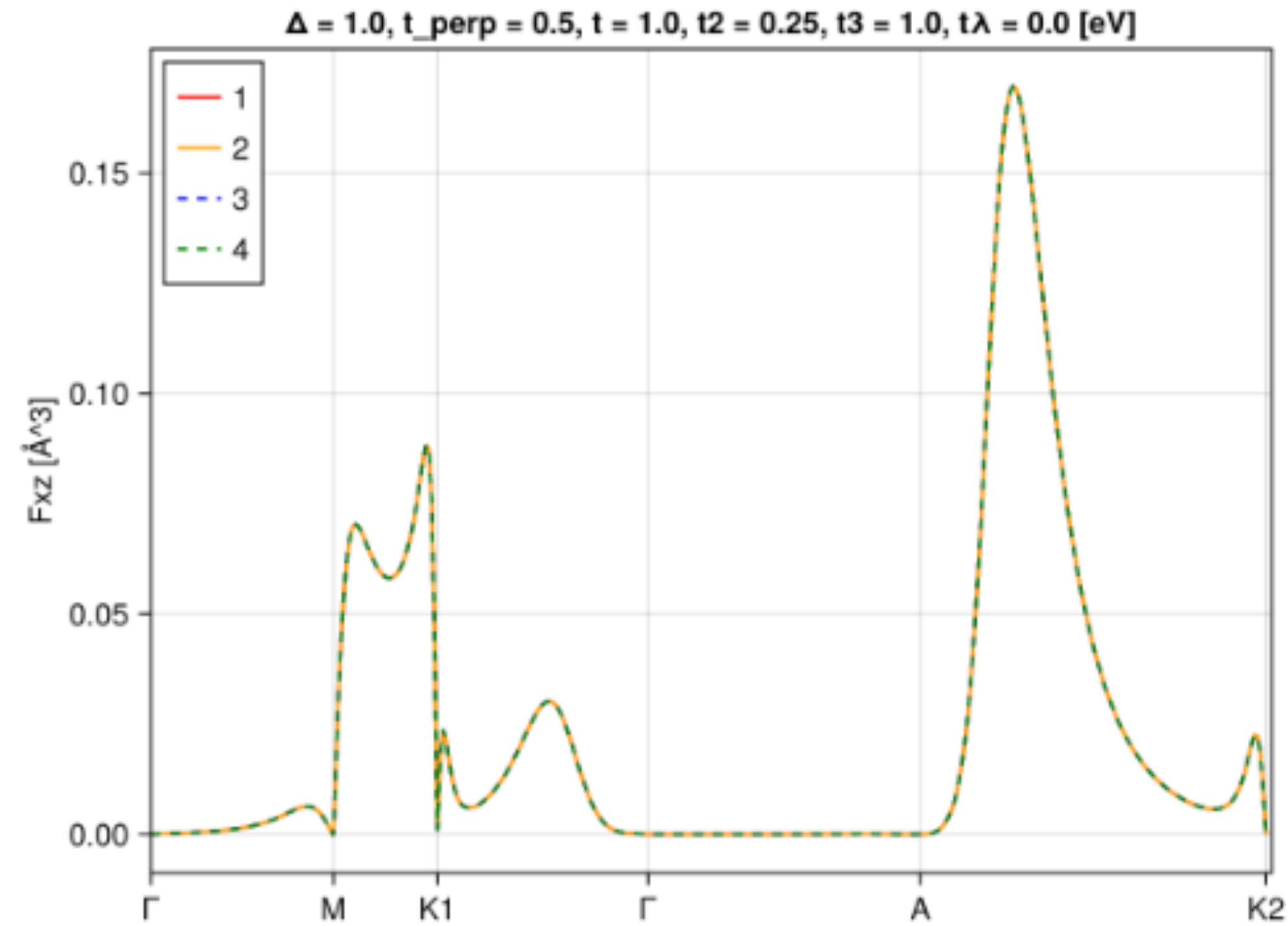
The spinfull ferroaxial model with KM SOC. Topological check

Adding a Δs_z



K-resolved calculations: Fxz

$$t_{\lambda} = 0$$



$$t_{\lambda} \neq 0$$

